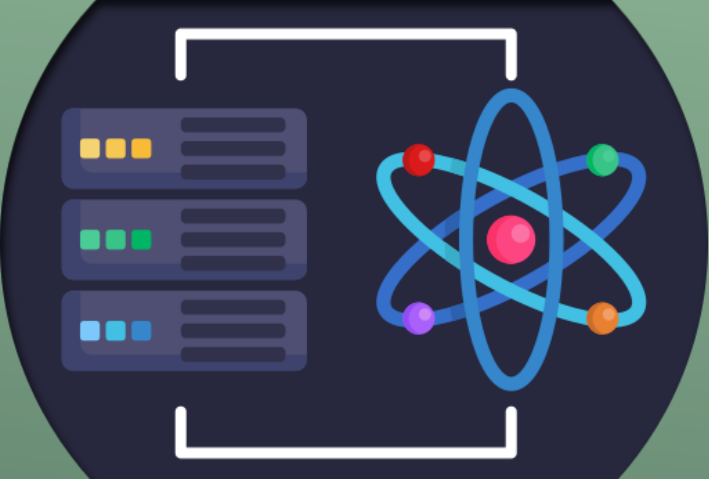


PREDICTING THE BLAZE: BRIDGING THE GAP BETWEEN PYTHON & ARCGIS

Jessica Nelson

Ethan Laundry



WHAT TO EXPECT

Introductions (a.k.a. building cred)

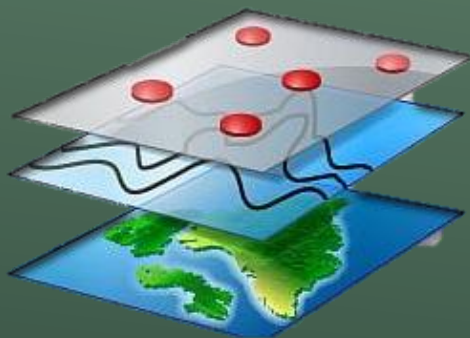
What's Python Programming

Python ArcGIS Bridge

Typical Use-Cases

Fairfax Script Tool Example

Questions (a.k.a. grill presenters)



WHO ARE WE

Jessica Nelson

- COVID class graduate 2020
- BS in criminal justice & biology
- 6 years of experience
 - 1 year as a police intern
 - 1 year as a DHHS data administrator
 - 4 years as a fire/EMS data analyst
- CPC Fire & Emergency Services Analyst (FESA) Commissioner
- Outreach Chair, Association of Fire & Emergency Services Analysts
- Alternate, NFPA's 1020 and 950/951 standards
- Recently married, #WifeyStatus

Ethan Laundry

- Graduated in 2022 with a BA in Economics
- MS in Statistics- Applied Statistics Concentration
- 3 years of experience
- 2 years as an actuarial data analyst
- 1 year as a fire/EMS data analyst
- FESA Applicant (in-progress)
- Outreach Committee member, Association of Fire & Emergency Services Analysts
- Likes to ride fire trucks, not ambulances
- Recently engaged (last month)

PYTHON INTRODUCTION (IT'S NOT SCARY)



CONNECTION BETWEEN PYTHON AND ARCGIS



ArcPy: python package

ArcGIS API for Python: web
GIS and admin

ArcGIS API for Python

- What can it do for you?

Platform API

**Power Users
Developers**

Enterprise Integration

Users, Roles & Group management



**Administrators
DevOps**

Spatial Analysis

Imagery

Location Analytics

Analysts

Big Data

Raster Analytics

Feature Analytics

Data Scientists

Content Publishers

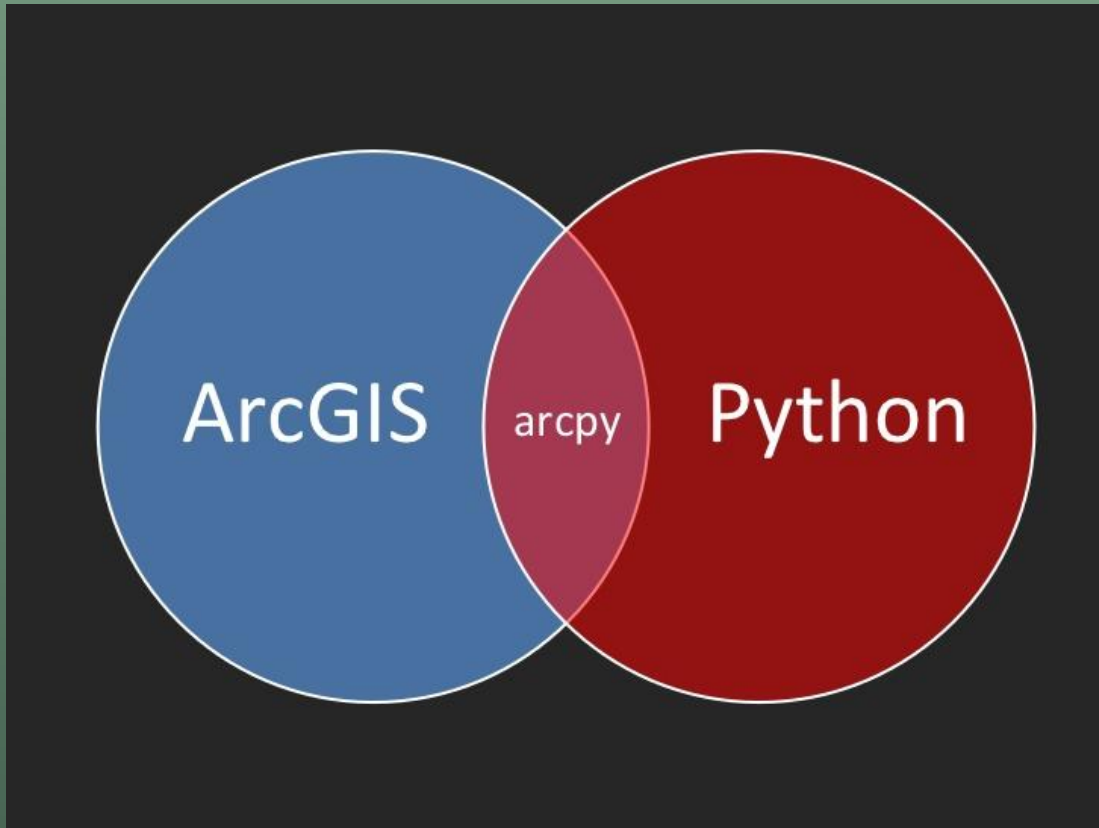
Your Web GIS

Script and automate your Web GIS

Data Management

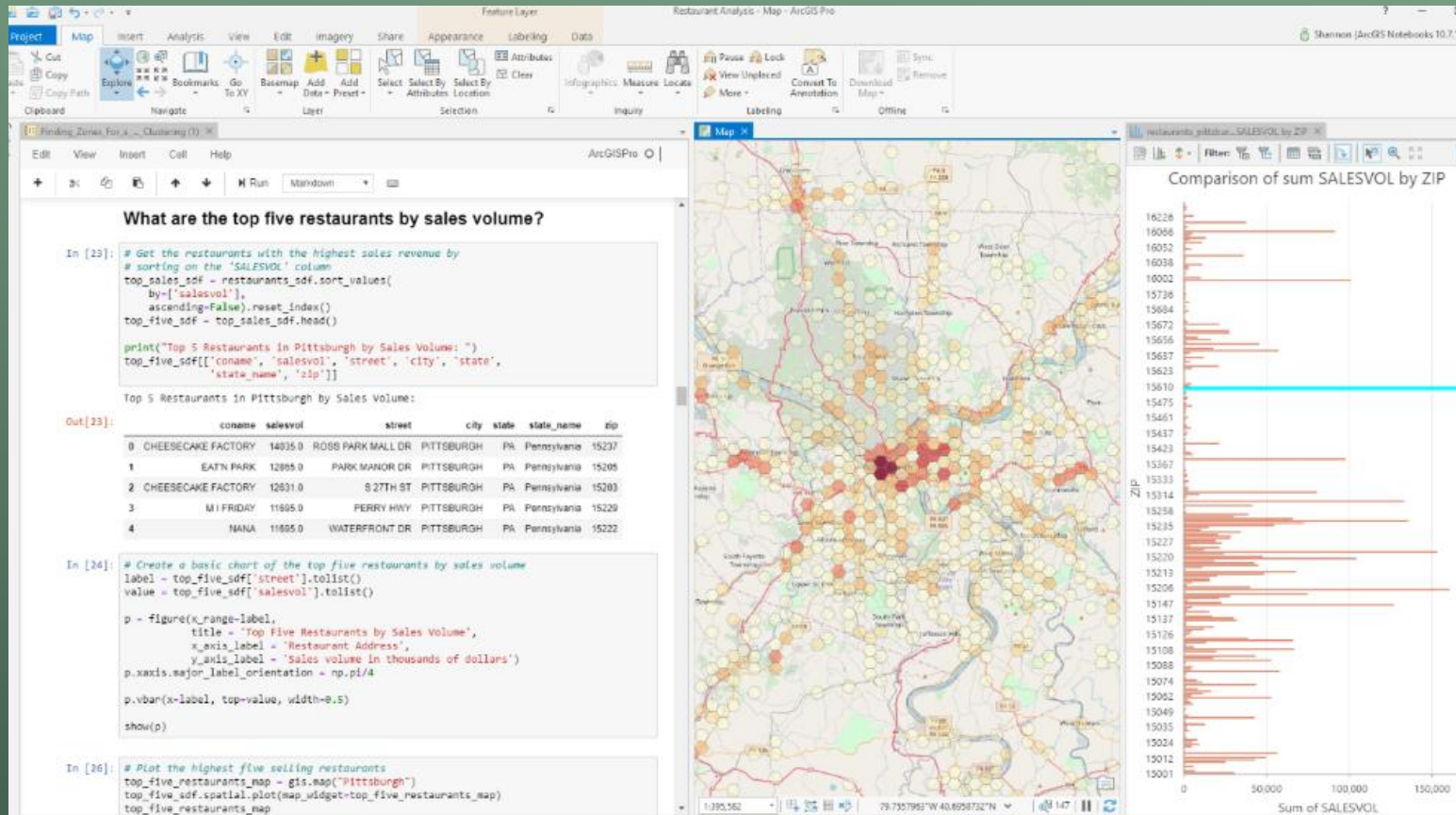


ESRI'S 'ARCPY' PYTHON PACKAGE



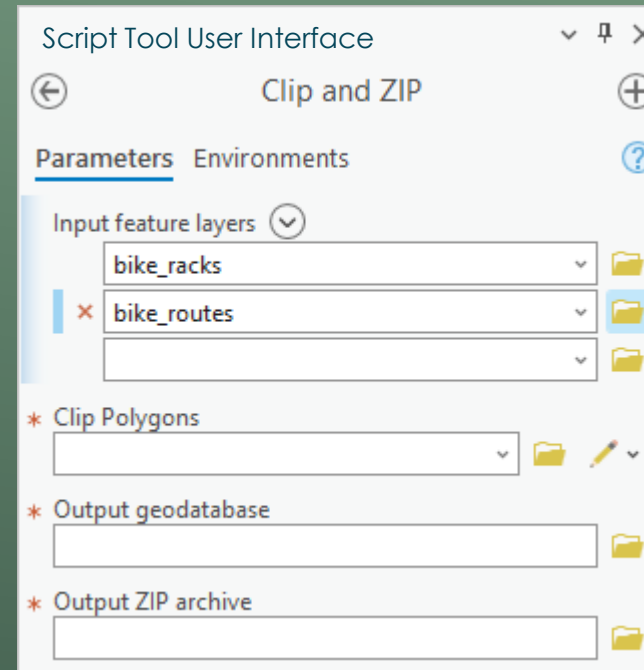
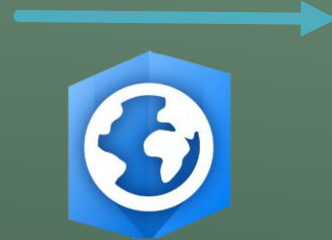
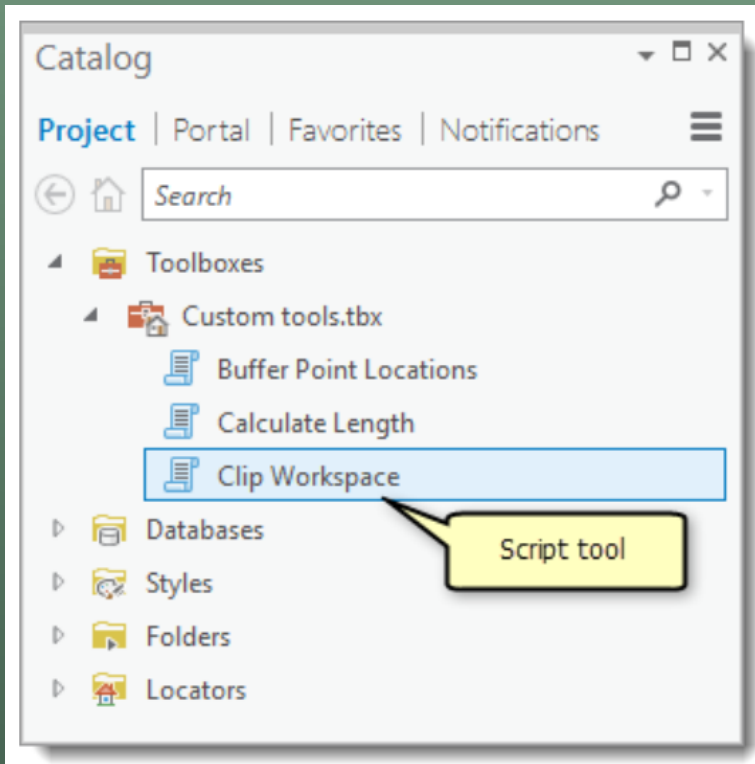
```
import arcgis  
import arcpy
```

ARCPY IN ARCGIS NOTEBOOKS



Python notebooks
are built into ArcPro

WHAT IS A SCRIPT TOOL



```
clip_zip.py - C:\Tutorials\PythonTool\clip_zip.py (3.11.8)
File Edit Format Run Options Window Help

import arcpy
import os
import zipfile

# Workspace and datasets
arcpy.env.workspace = "C:/Tutorials/PythonTool/DC.gdb" # Geodatabase wh
clip_fc = "C:/Tutorials/PythonTool/neighborhood.shp" # Full path of c
gdb = "C:/Tutorials/PythonTool/Clip.gdb" # Full path of o
gdb_path, new_gdb = os.path.split(gdb) # Split full path
out_zip = "C:/Tutorials/PythonTool/Clip.zip" # Full path of t
arcpy.env.overwriteOutput = True

# Create a new geodatabase to store the result
arcpy.CreateFileGDB_management(gdb_path, new_gdb)
print(f"Output geodatabase {gdb} created")

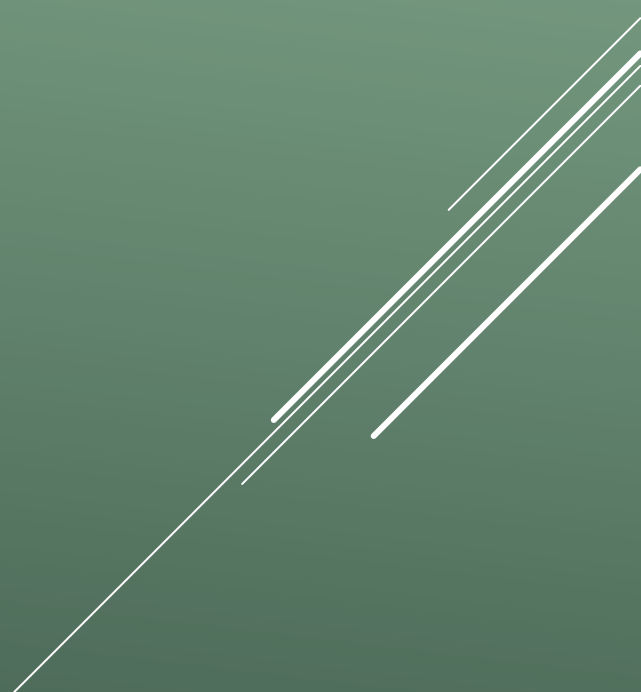
# Clip each input feature class
inputs = arcpy.ListFeatureClasses()
print(inputs)
for fc in inputs:
    fc_name = arcpy.da.Describe(fc)["baseName"] # Remove folder an
```



BASIC ARCPY: SCRIPT TOOL

➤ Benefits:

- ✓ Customizable
- ✓ Standardize
- ✓ Reproducible
- ✓ Repeatable, monthly execution
- ✓ Parameterized inputs (Filters)



BASIC ARCPY: SCRIPT TOOL

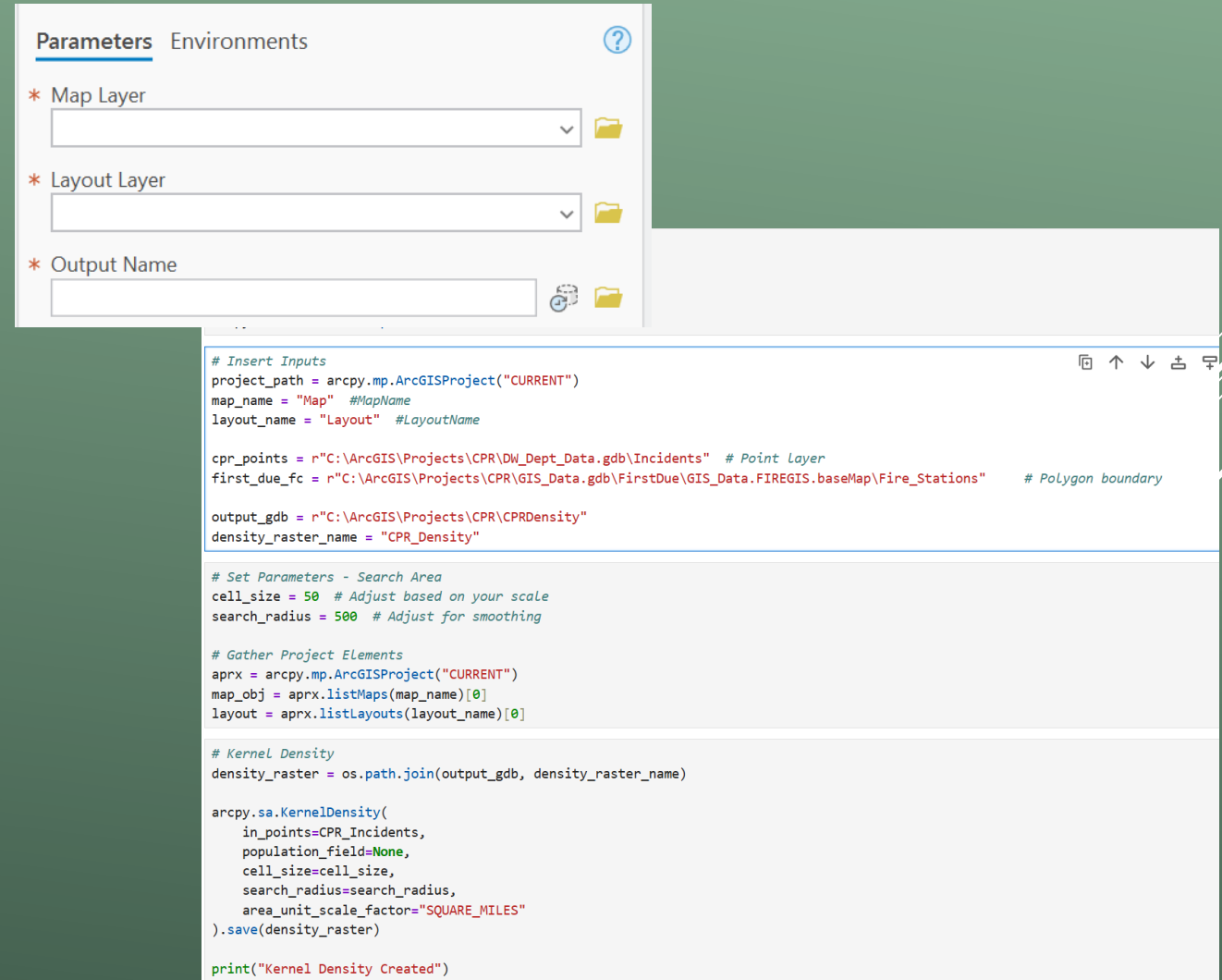
- ▶ What's the question are you trying to answer?
 - ▶ Some questions that could be answered by a script tool:
 - ▶ What is the impact of a 15-minute delay on mutual aid Rescue Company dispatches?
 - ▶ What area/how far is it likely for a wildfire to spread?
 - ▶ What areas fall outside of 4-, 8-, or 12-minute travel time coverage?
 - ▶ What communities should CRR target for CPR training?
 - ▶ What's the impact of the surrounding first dues if we convert a station to peak staffing?
- 
- A series of three parallel white lines of varying lengths, slanted diagonally upwards from left to right, located in the bottom right corner of the slide.

BASIC ARCPY EXAMPLE: SCRIPT TOOL

- ▶ Example: Where are the CPR incidents in a specific first due?

User: selects area of interest & date from script tool UI. Hits run.

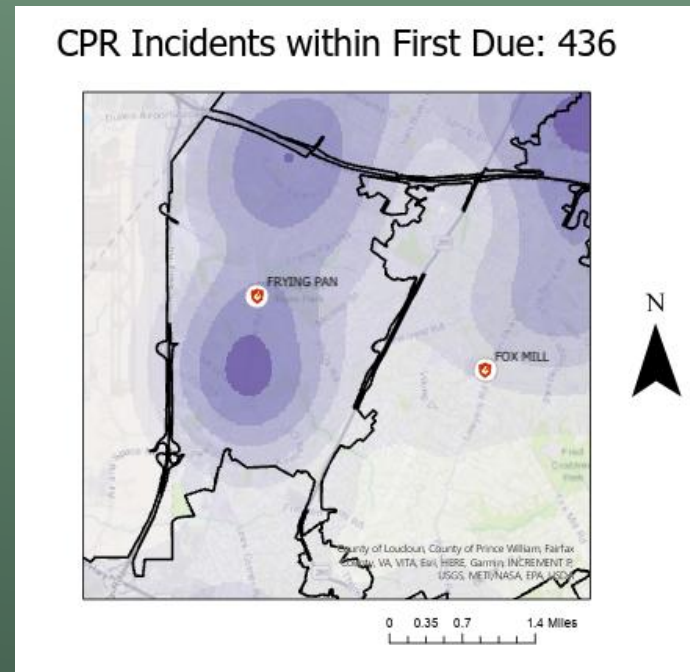
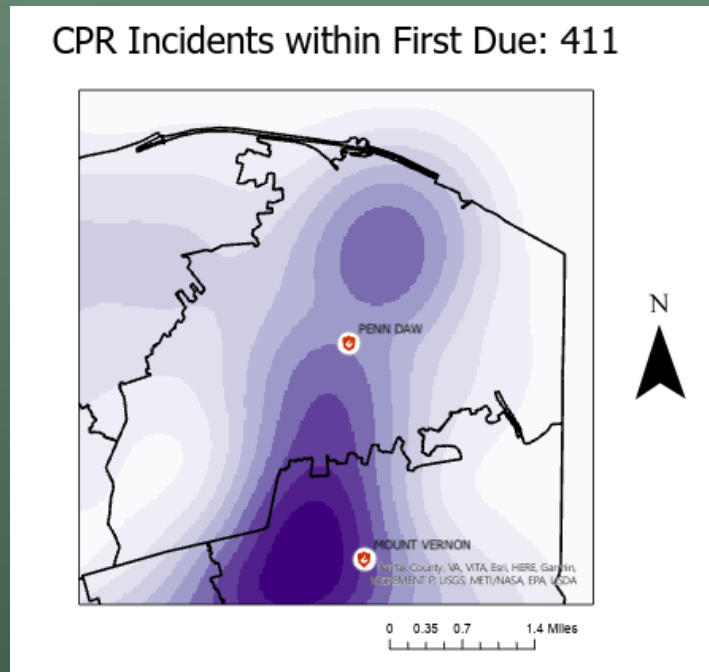
BTS: Script checks incidents in the county and creates a density map of the areas where a CPR incident had occurred



BASIC ARCPY EXAMPLE: SCRIPT TOOL

- Example: Where are the CPR incidents in a specific fire due?

Output: Map CPR incidents spread



BEYOND MAPS-- CHARTS



RELATIONSHIPS

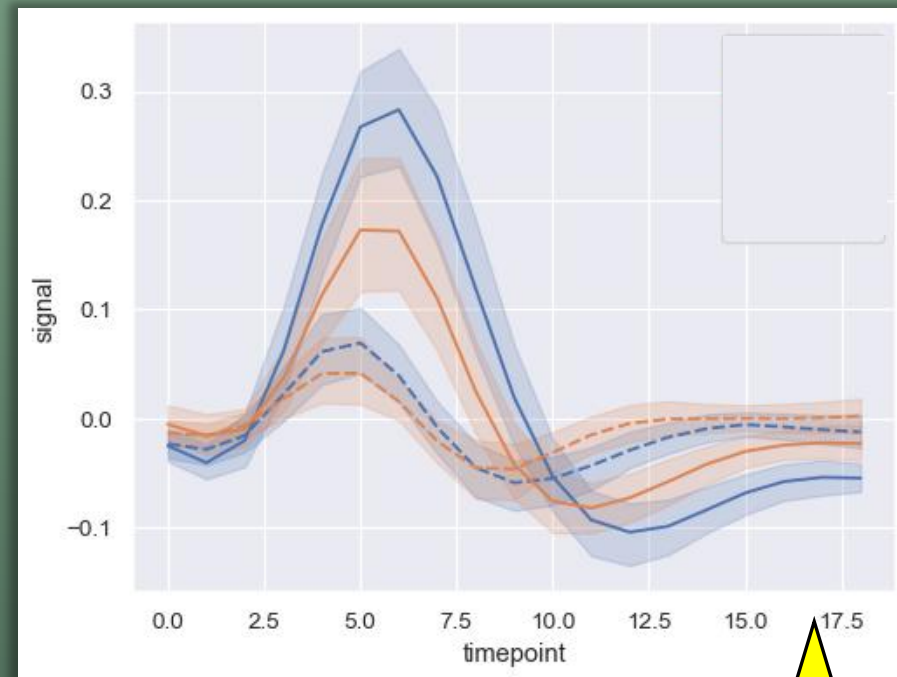
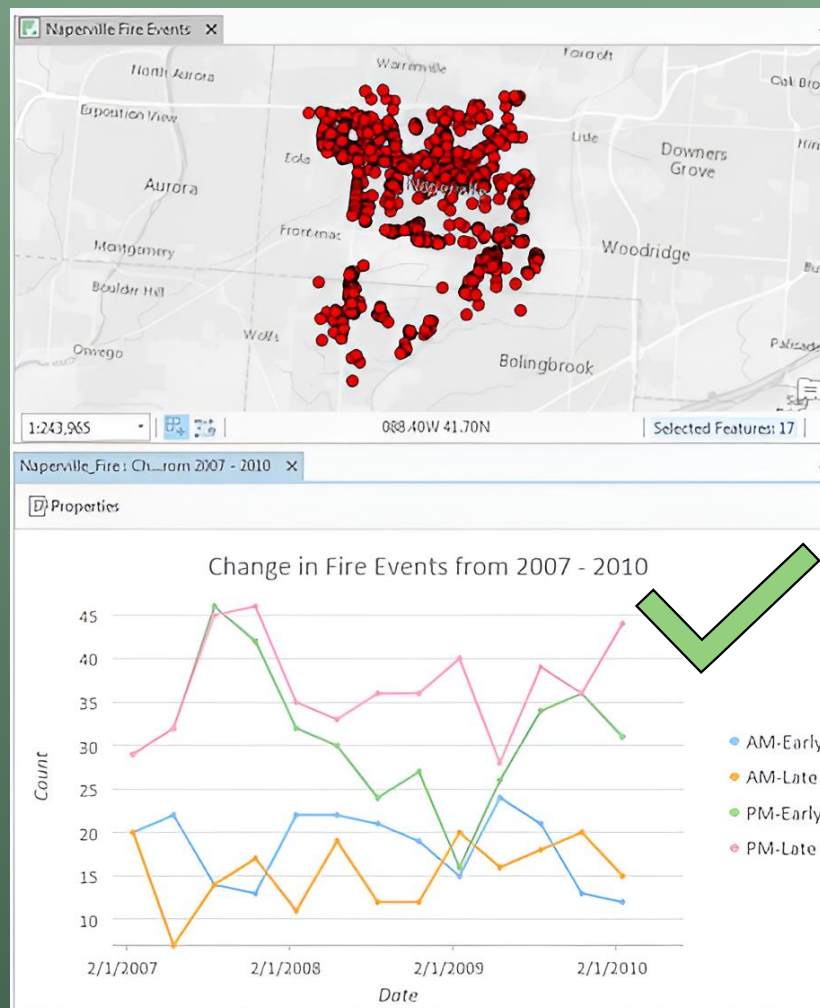
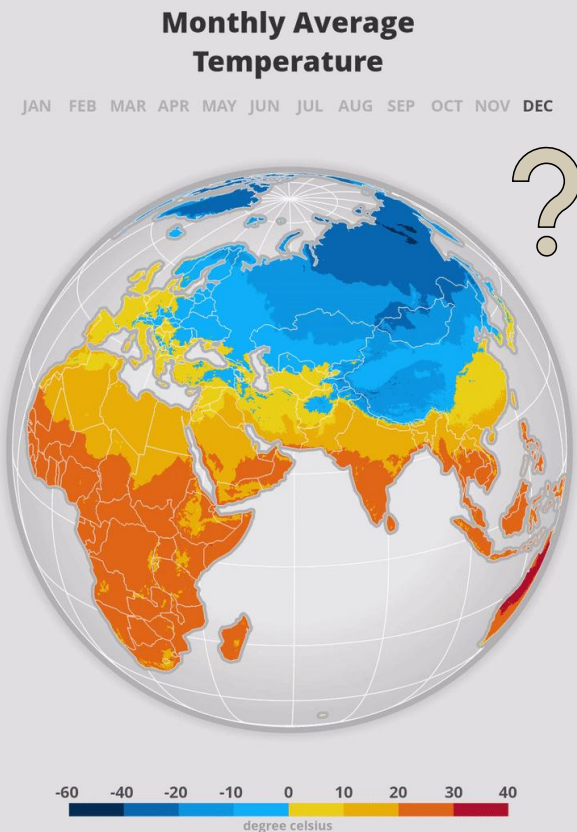


CHANGE-OVER-TIME

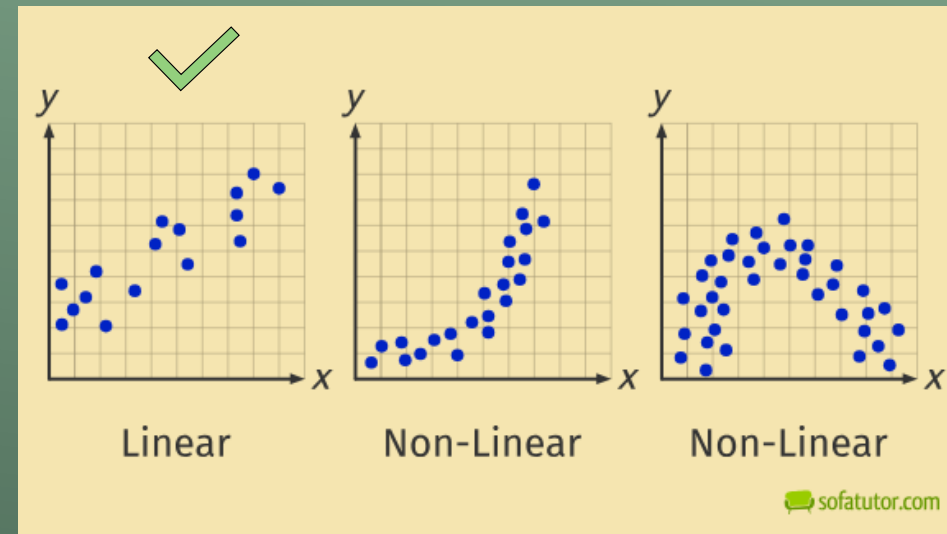
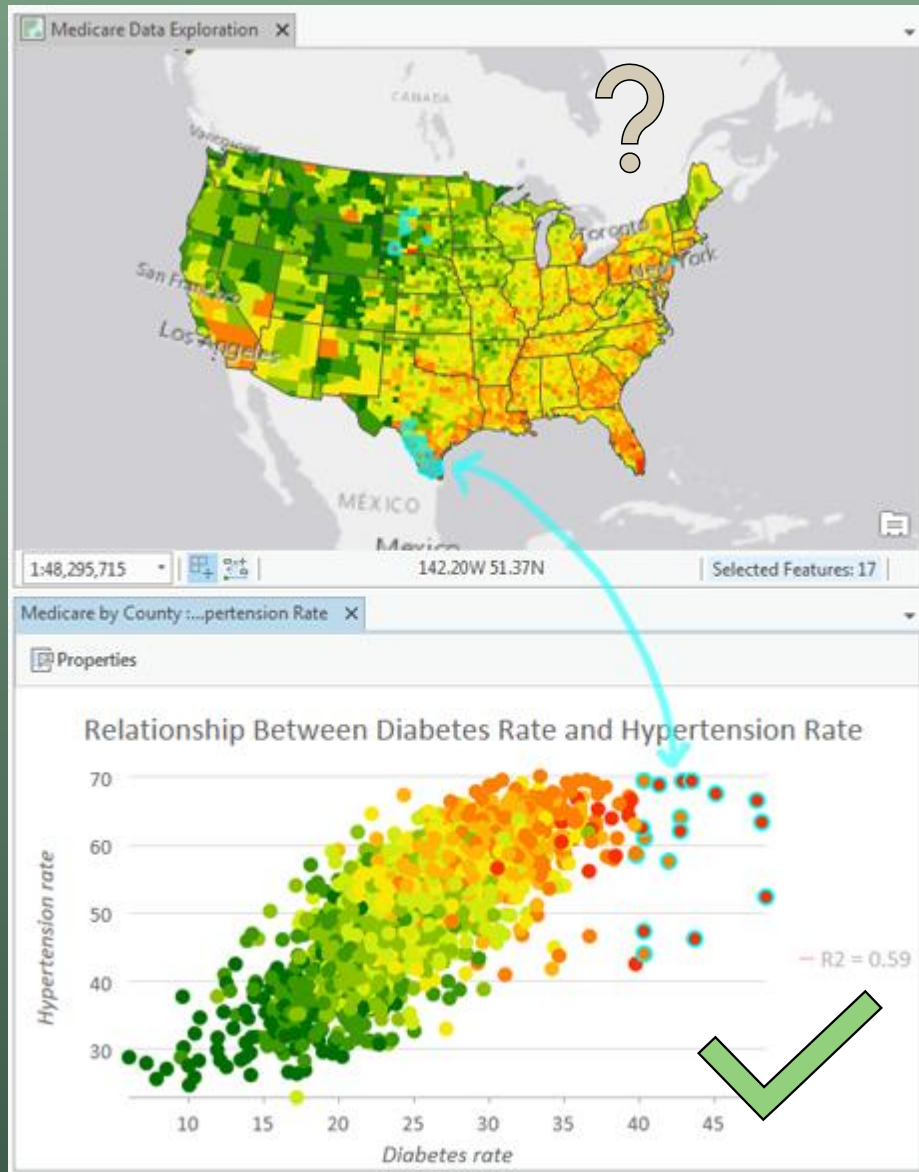


EXAMPLE: CHANGE-OVER-TIME

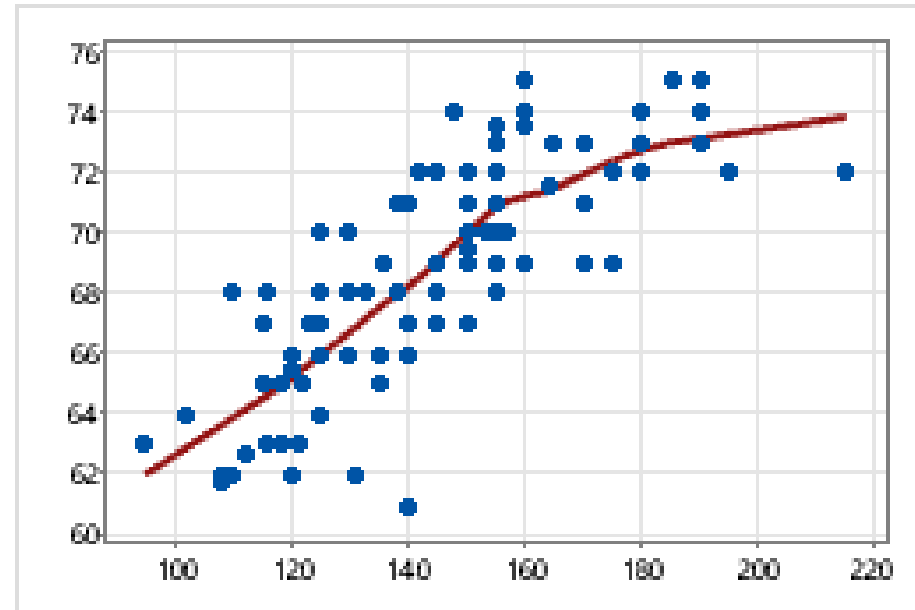
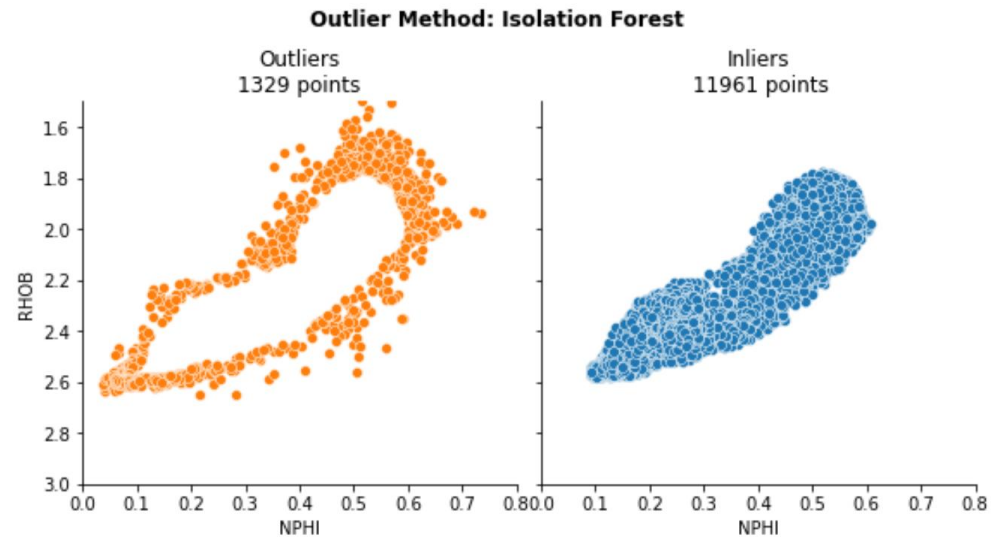
Package: seaborn



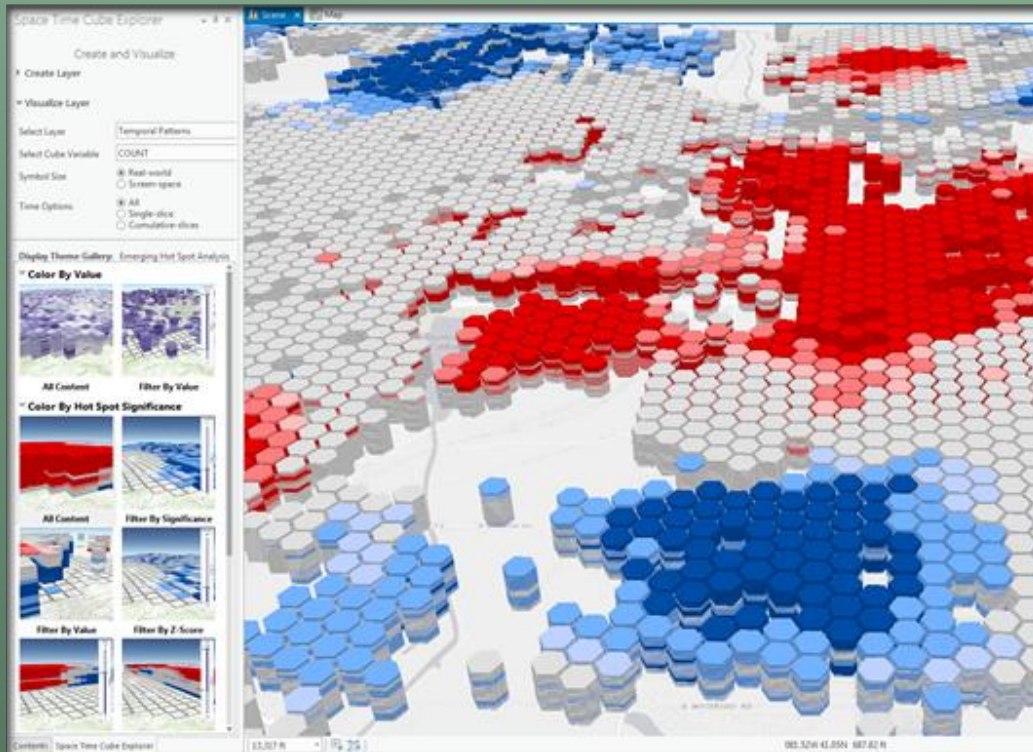
EXAMPLE 2: SCATTER PLOT



Outlier Method: Isolation Forest
Number of anomalous values 1329
Number of non anomalous values 11961
Total Number of Values: 13290



EXAMPLE 2: SCATTER PLOT COT'D



- ▶ Custom script tool to model and simulate response times ★
- ▶ Use (and/or replicate) Esri's premade spatial statistics tools
- ▶ Hide complicated back-end code behind User Interfaces (UI)

HOW IS FAIRFAX USING PYTHON WITH GIS

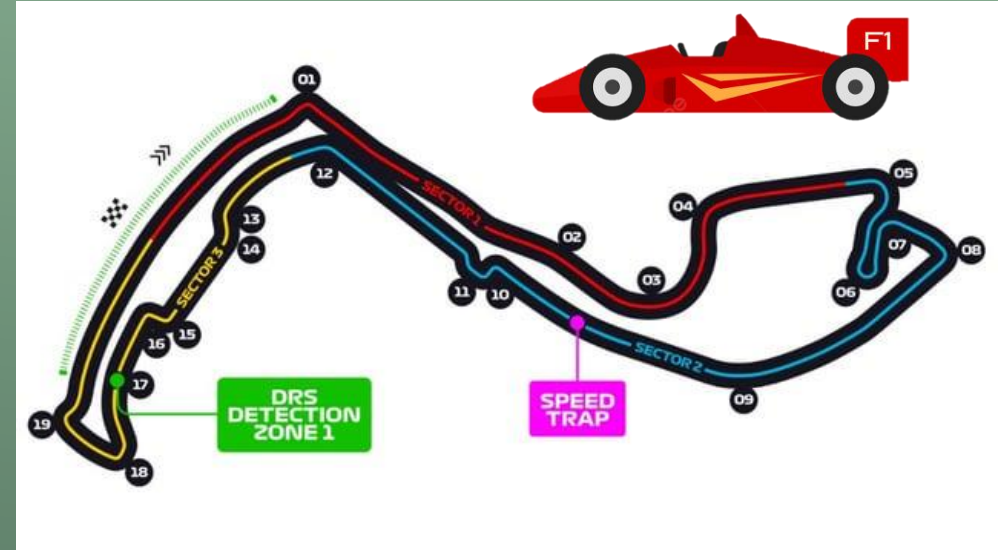
EXAMPLE: F1 RACE STRATEGY

Goal: Win the race

Fixed Inputs: track layout, car design

Emergent Inputs: weather conditions, accidents

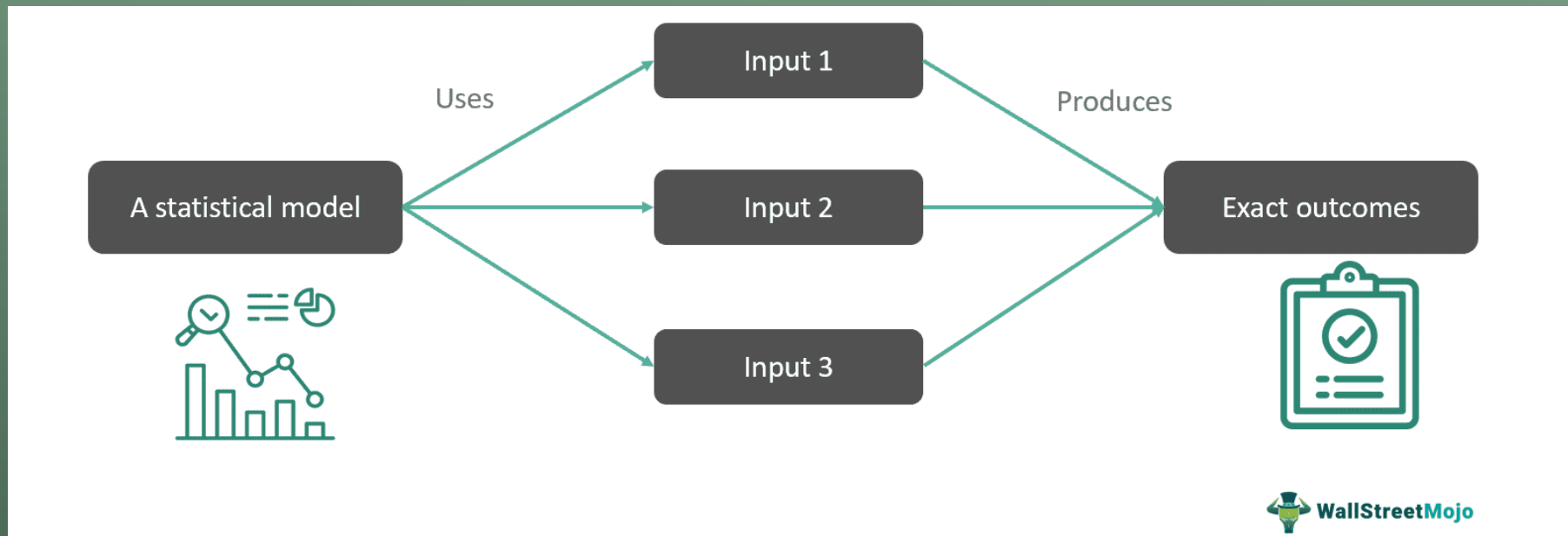
Decision Variables: pit stop timing, tire choice, overtaking attempts



FORMULA 1 EMILIA ROMAGNA GRAND PRIX 2024					
RACE RESULTS					
P2 NORRIS		P1 VERSTAPPEN		P3 LECLERC	
+0.725	18 PTS	1:25:25.252	25 PTS	+7.916	15 PTS

MOTIVATION EXAMPLE: F1 RACE STRATEGY

Question: Given the uncertainties in race conditions, can deterministic models reliably predict winning strategies?



Answer: No! They fail to account for random events and variability

MOTIVATION EXAMPLE: F1 RACE STRATEGY

Solution: Use Monte Carlo simulation (in Python, etc.) to model uncertainty and evaluate strategy across many scenarios



MOTIVATION EXAMPLE: F1 RACE STRATEGY

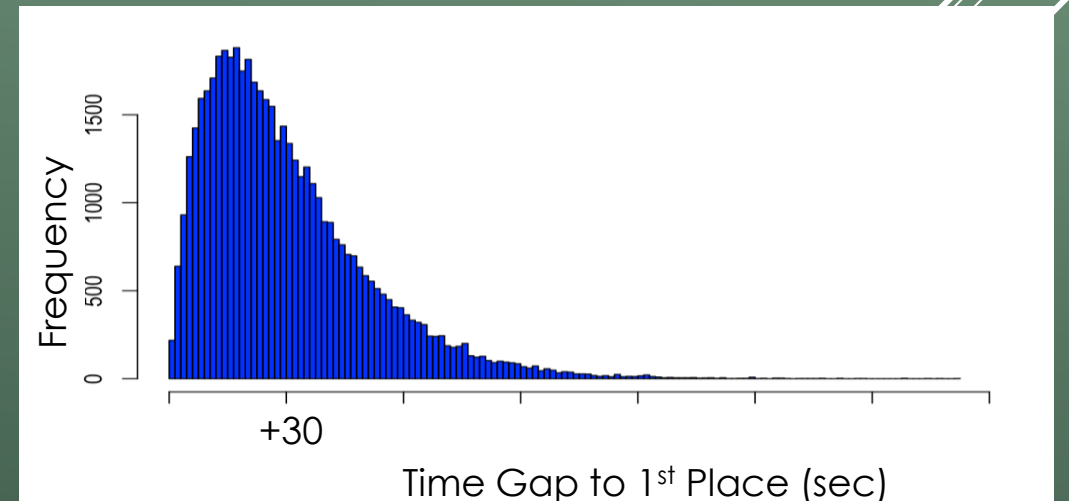
Thousands of simulations give teams **probabilities** to answer key questions:

Should we take a pit stop now?

- Pitting now → 38% win probability
- Pitting in 3 laps → 29% win probability

What's the risk my tires give out soon?

- 27% probability of severe tire degradation within the next 8 laps



FCFRD RESPONSE TIME SIMULATION

- What's the probability of a 6-minute response to surrounding areas, given a proposed staffing change?

Are we at least 90% confident we can meet that benchmark?

What about during snowy conditions?

Monte Carlo simulation is the gold standard for optimizing decisions under uncertainty — in F1 racing or emergency response

FCFRD RESPONSE TIME SIMULATION

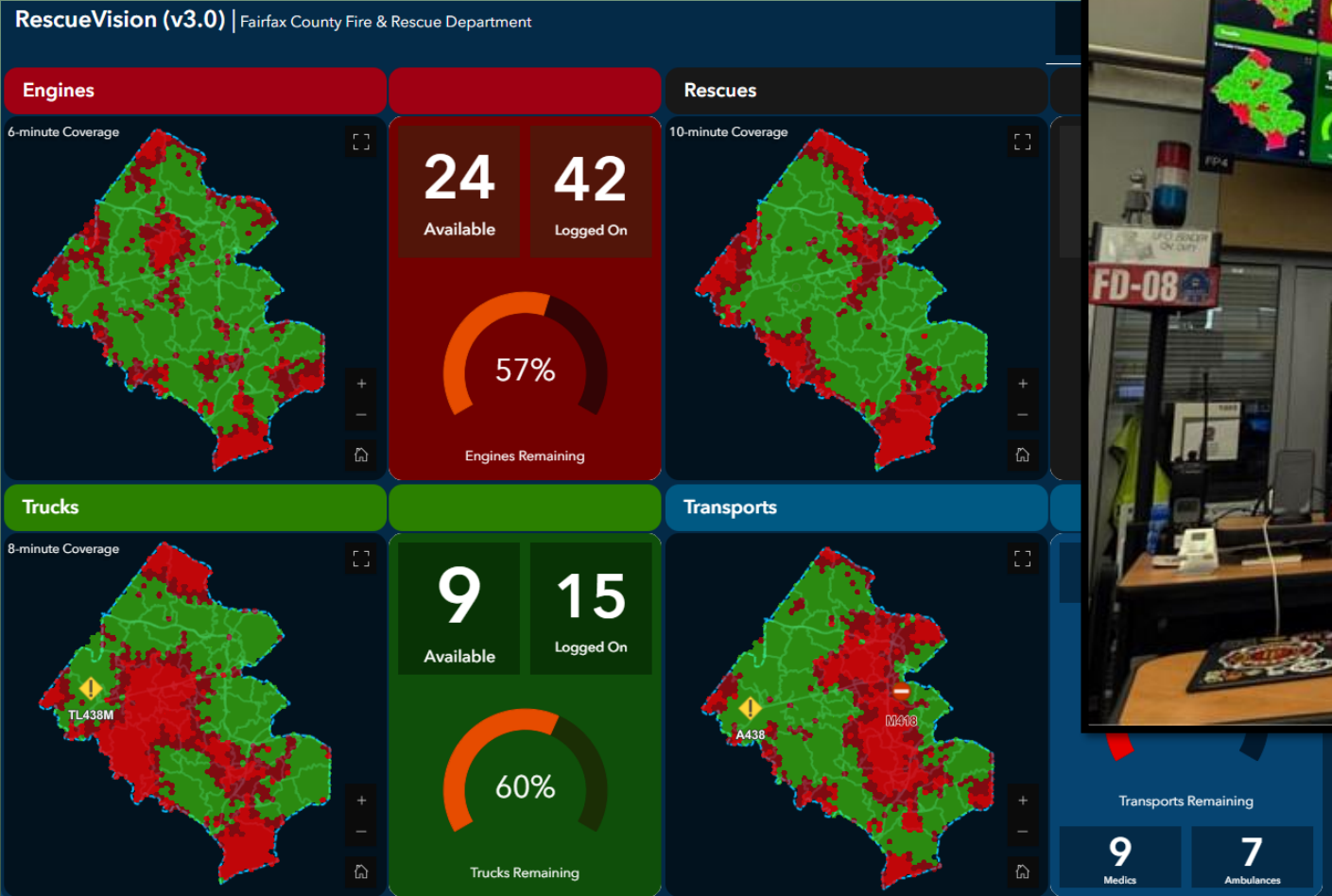
Goal: pre-simulate thousands of possible scenarios/outcomes for certain decisions—know the optimal strategy up-front!

Use Cases:

- Optimize Staffing Locations
 - Pre-plan Uncommon or Complex Scenarios
 - Anticipate Coverage Gaps & Percentile Outcomes
 - Assist Real-Time Dispatch Strategy
- 
- A series of four parallel white lines of varying lengths, slanted diagonally upwards from left to right, located in the bottom right corner of the slide.

FCFRD RESPONSE TIME SIMULATION

Rescue Vision



FCFRD RESPONSE TIME SIMULATION

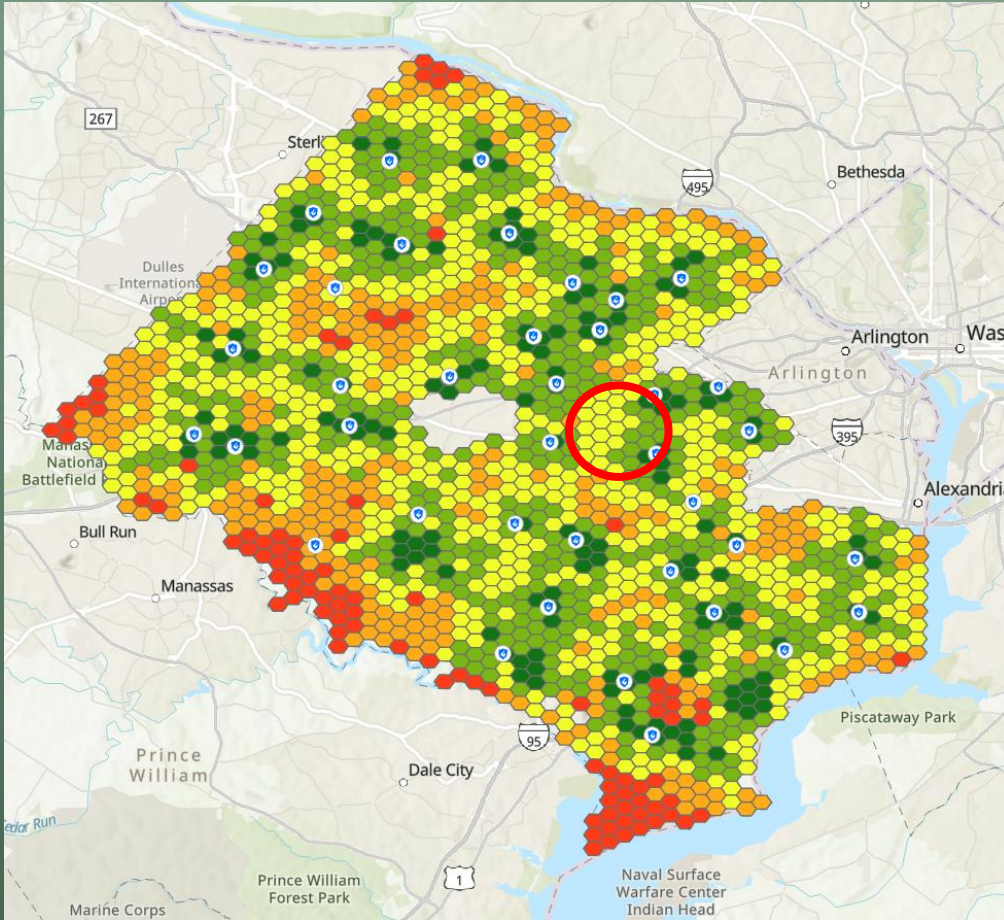
And because it's a script tool it's:

- ▶ Easy to Use
- ▶ Validation
- ▶ Repeatable
- ▶ Works directly with ArcGIS mapping

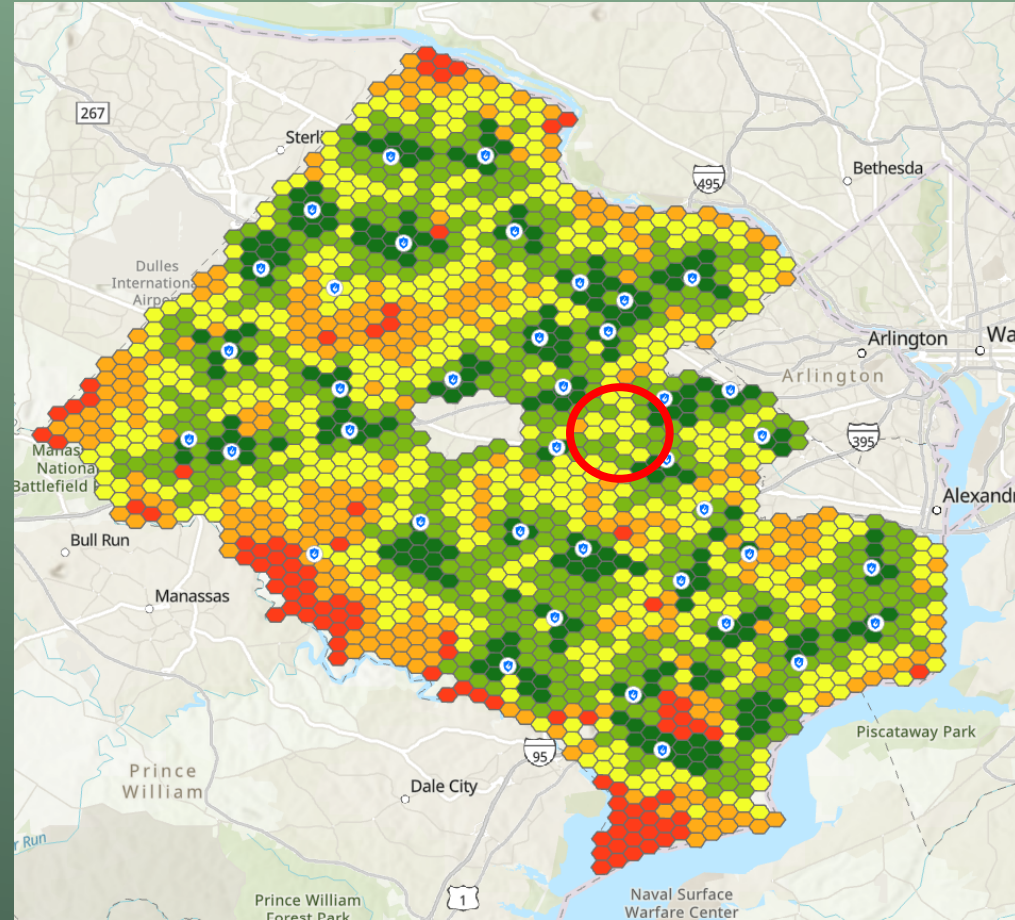
The screenshot shows the 'Geoprocessing' window for the tool 'Pull ResponseTime Simulations'. The 'Parameters' tab is active. The interface includes the following elements:

- Time Window of Interest:** A dropdown menu.
- Unit Type(s):** A dropdown menu.
- Indicate Brown Outs (optional):** A section with three input fields:
 - Unit Type (Ambulance, Truck, etc.)
 - Station (401, 402, etc.)
 - # to Brown Out (1, 2, etc.)
- Buttons:** An 'Add another' button with a plus icon.
- Options:** Two checkboxes at the bottom:
 - ☐ Exclude Call Processing Times
 - ☐ Write Outputs to SQL

FCFRD RESPONSE TIME SIMULATION



6 am – 9am
(Morning Rush)



3 pm- 5 pm
(Evening Rush)

- Transport Units Availability
- 495 (highway impact)

QUESTIONS, THOUGHTS, FEELINGS....



THANK YOU & RESOURCES



Jessica Handoko Nelson, FESA



Amadea.Nelson@fairfaxcounty.gov



571-567-1569



<https://www.linkedin.com/in/ajhandoko>



Ethan Laundry



Ethan.Laundry@fairfaxcounty.gov



571-992-3150



<https://www.linkedin.com/in/ethan-laundry/>

- ▶ [ArcPy Essentials | Learning Plan](#)
- ▶ [Get started with Notebooks in ArcGIS Pro](#)
- ▶ [ArcGIS Pro Python reference—ArcGIS Pro | Documentation](#)